Forms of Reduction: Affordances of Tokenization Algorithms in Neural Machine Translation

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ABSTRACT

This study offers a genealogical and technical account of opacity in neural machine translation (NMT), situating contemporary large language models within a longer history of attempts to formalize language as computation. It traces this history to early machine translation and communication theory, where language was conceptualized as separable from a language-invariant computational substrate. Within this framework, linguistic input and output function as interfaces to an abstract computational domain, obscuring the operations through which language is transformed. The thesis examines tokenization as a key site at which this logic becomes operational. By decomposing language into statistically defined units, tokenization reconstitutes the unit of linguistic meaning in computational rather than linguistic terms. A comparison of Byte-Pair Encoding and Unigram demonstrates how different algorithms formalize and manage redundancy, producing distinct subword representations while concealing the operations through which those representations are generated. The analysis then extends this problem of opacity to natural language generation, arguing that these systems do not simply produce opaque outputs but also shape the conditions under which their operations can be understood. By privileging textual interfaces, natural language generation systems make their underlying computational processes difficult to perceive or interrogate. Tokenization therefore emerges not merely as a technical preprocessing procedure, but as a cultural and epistemological mechanism through which language, meaning, and agency are reconfigured in computational contexts.

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INTRODUCTION

The term "black box" is commonly thrown around in the discussions of large language models, both public discourse and academic fields, to point at LLMs' opacity. When interacting with an LLM, users encounter fluent and coherent language, but the processes that produce this output are intransparent. Having asked an everyday question and received a convincing answer, the interface might begin to feel all too human: if it quacks like a human, it might as well be, to some part, human; users attribute understanding or intention to the system because its operations are hidden. As Peter-Paul Verbeek and Don Ihde suggest, this reflects a tendency to treat technologies as "quasi-others" when their functioning is not directly observable:[1] a form of projection of human consciousness to things completely external to the human social and physical body.

A large language model is a system trained on large amounts of text to reproduce statistical patterns in language. It does not store meanings or grammatical rules; instead, it predicts the next token in a sequence based on learned probabilities. Although this task is narrow, it produces outputs that resemble understanding when scaled up.

As these systems have become embedded in everyday tools, their internal complexity has been increasingly framed as opaque, and the vocabulary of the "black box" has migrated from theoretical discourse into broader cultural discussion. Karasinski and Candiotto observe that the "black box" image casts AI as an "unfathomable power ... shrouded in an aura of opacity," a perception they argue is partly mythical. This narrative of the black box finds its place in industry discourse, too: IBM's explainable-AI (XAI) page warns that advanced AI's "whole calculation process is ... a 'black box' that is impossible to interpret".[2] The opacity in question can be understood as a technical property of LLMs insofar as it follows directly from the conditions under which such systems are made to function. The system is fully specified—in a limited thought experiment, one might, theoretically, inspect every parameter and gradient—but the model comprises billions of parameters whose interactions are highly nonlinear and context-dependent, making it infeasible to

[1] Richard Heersmink et al., "A Phenomenology and Epistemology of Large Language Models: Transparency, Trust, and Trustworthiness," *Ethics and Information Technology* 26, no. 3 (2024): 52, 52, https://doi.org/10.1007/s10676-024-09777-3.

[2] Murilo Karasinski and Kleber Bez Birolo Candiotto, "AI's Black Box and the Supremacy of Standards," *Filosofia Unisinos* 25, no. 1 (2024): 1, https://doi.org/10.4013/fsu.2024.251.13.

derive an account of how a particular output is produced in practice.

There is a sense in which the "black box" is plastered over indifference both in research and in popular contexts. In popular discourse, the same term circulates as a ready-made explanation, absorbing anxiety or fascination without requiring further specification of different modes of absence—the unknown, the unknowable, the unexamined. In research contexts, it can function as a placeholder that permits continued emphasis on performance gains while deferring harder questions about robustness and failure modes; the opacity is acknowledged, named, and then operationally bracketed. Perhaps, this is what makes black box one rare tool of science and technology studies that has a name—as it is a form of delineation and simultaneous disregard of the opaque.

I open this work with the 'black box' as it circulates in popular and public discourse, because the term has become redundant there: a ready-made label that absorbs both anxiety and fascination while requiring no further specification of the different modes of absence it names. This suggests that different technical explanations may lead to the same cultural perception of opacity. The question, then, is whether the specific technical reason for opacity matters for how the "black box" is understood and used—or whether it is enough that opacity can always be assigned some technically-specifiable form. While the cultural trope of the black box may remain indifferent to these distinctions, the particular mechanism shapes both the practical limits of interpretability and the historical conditions under which the opacity first became productive. It is this specificity that the analysis recovers by turning to the history of neural machine translation, the site where these technical commitments were first crystallized and where the relation between language, computation, and opacity was first engineered at scale.

0.1 The Argument

The 'black box' is one among many formulations of our inability to internalize technology—to describe it in full in human terms. Technology resists being fully internalized, being explained in human terms; stemming from outside of us, it creates a persistent challenge for theorists attempting to grasp it within traditional discursive frameworks; the 'black box' is just one formulation of this issue. Mark B. N. Hansen in his Embodying Technesis notices this tendency across different post-structuralist philosophies of technology. He notices two acts of reduction underlying the study of technology—the reduction of technology to machine and the

reduction of machine to text—and for the totality of this logical procedure Hansen coins the term technesis, the putting-into-discourse of technology.[3] This double reduction highlights how theory often approaches technology only through familiar conceptual categories or metaphors—in Hansen's analysis these are examples of poststructuralist thought: "Heidegger's horizon of thought, Derrida's closure of representation, Freud's psychic space, Lacan's objet a, and Deleuze and Guattari's social assemblage"[4]—which may obscure the material and affective dimensions of technology. Finding such a reduction in contemporary theory of technology puts at question the relevance of the very theory: it allows us to ask what is actually the object of the study of technology?—implying that the object that we have been talking of has never been technology itself; reduction is theory's pharmakon: it allows us to talk technology, but deprives us of technology's particularity.

I seek to extend this insight into STS and media studies to track if reduction is a methodological limitation or a technological affordance when it comes to machines that do operate with text. By stabilizing technology through procedures such as foregrounding cultural techniques as symbolic operations, redescribing mediation as rules of selection/storage/transmission, analyzing artifacts primarily in terms of their role in social order and epistemic authority, these fields render technology legible while systematically excluding its material and affective dimensions. Yet with large language models, this reduction is no longer merely a limitation imposed by scholarly methods—LLMs materially enact and afford the very logic of technesis. What scholars in media studies and STS have long performed discursively—reducing technology to its textual or symbolic trace—is here materialized as the natural language generation technology's mode of functioning: their organization, built upon a historical and operational separation between observable linguistic surface (input/output) and a computational substrate operating in non-linguistic units.

Across the three chapters, I ask and seek to answer three questions.

First, how did early machine translation research, communication theory, and their overlapping metaphors produce the conditions for today's black-box opacity, making the path between input and output structurally arbitrary yet functionally convenient? Thus, in Chapter I I argue that the distinctive black-box opacity of LLMs occurs in the historically specific separation produced by the metaphors of early MT: the treatment of linguistic surface forms (input and output) as detachable from

[3] Mark B. Hansen, *Embodying Technesis: Technology beyond Writing* (Ann Arbor: University of Michigan Press, 2000), 4.

[4] Hansen, *Embodying Technesis: Technology beyond Writing*, 20.

an underlying computational invariant layer. By tracing the emergence of this separation in mid-twentieth-century machine translation research—particularly in the communication-theoretical framework of Claude Shannon and Warren Weaver, and most explicitly in Weaver's 1949 tower-and-basement metaphor of the founding Translation memorandum—I show how language generation technologies came to be imagined and engineered in a split between the input/output layer and internal operations.

If Chapter I establishes the historical conditions of opacity, in Chapter II I look at the technical grounds of this obscuring, zooming into the transition between the input layer and internal operations of contemporary neural machine translation (or NMT). My task there is to isolate the precise technical site in NMT where the black-box opacity later generalized to LLMs is first materially produced. This leads me to address the particularities of tokenization, computational encoding of natural language inputs. Therefore, my second question is how does tokenization—the foundational act of inscribing natural language into computable form—displace linguistic units of meaning and obscure the distinct ways different algorithms handle the redundancy of language?—which I address in a short comparative reading of Byte-Pair Encoding (BPE) and Unigram tokenization algorithms, which present greedy compression versus probabilistic embrace of redundancy, respectively. The payoff of this analysis is to show that the opacity of contemporary LLMs begins not as a feature of scale but as the outcome of these encoding decisions first crystallized in machine translation.

The last part of my work is dedicated to the methodological stakes that follow—in that the first two chapters establish that reduction to textual inputs and outputs is historical and structural, not methodological. What are the epistemic consequences for media studies and STS when the technology under analysis actively enforces reduction to text as its mode of operation, rendering its non-linguistic substrate constitutively illegible, while simultaneously sustaining a seamlessly textual interface? In this case, when scholars stabilize LLMs through familiar lenses—such as sociomaterial assemblages, cultural techniques, sites of power relations—they risk aligning with the very logic the systems embed. Such approaches treat the textual interface as the primary site of analysis, thereby reproducing the separation the architecture already enforces. The challenge is therefore not to overcome opacity through better metaphors or more nuanced social contextualization, but to recognize how certain material configurations systematically privilege textual legibility

while rendering non-discursive dimensions illegible. LLMs enact the logic of technesis—they structurally solicit and reward reduction to text, shaping the kinds of questions and interpretations that appear natural or productive. Thus the analysis closes the circle opened by the black box: the constitutive opacity traced from its historical and technical genesis in NMT is not an external puzzle to be solved but the very condition that structures how language becomes computable, requiring media studies and STS to rethink their own analytic habits rather than reproduce the textual reduction the architecture already enforces.

0.2 The Object

To locate the symptoms of the particular opacities of LLMs, it is necessary to establish the greater object of this study defined in the title: neural machine translation. Large language models and machine translation share a common technical and historical development. Machine translation is particularly useful for this study because many of the key problems of contemporary LLMs were first formulated there. In Ataman et al., "Machine Translation in the Era of Large Language Models: A Survey of Historical and Emerging Problems," machine translation is described as the site where many of the core problems of today's LLMs were first articulated. Early approaches framed translation as a problem of mapping words or structures between languages; statistical methods reformulated it as estimating probabilities from large collections of parallel texts; neural approaches then reconceived it as learning continuous representations capable of generating entire sentences.[5]

For instance, the Transformer architecture, which now underlies most LLMs, emerges directly from this trajectory: it was originally proposed to address limitations in earlier translation models, particularly their difficulty in capturing long-distance relationships and their reliance on sequential processing.[6] The Transformer made it possible to model all parts of a sentence in parallel while still tracking their interrelations; although no longer confined to translation tasks, this architecture retains the same fundamental objective: to model relationships within and across part of natural language sequences. Machine translation also developed key data practices. The construction of parallel corpora—aligned texts in two or more languages—provided not only the scale of data required for training but also a concrete operationalization

[5] Duygu Ataman et al., "Machine Translation in the Era of Large Language Models: A Survey of Historical and Emerging Problems," *Information* 16, no. 9 (2025): 723, https://doi.org/10.3390/info16090723.

[6] Ashish Vaswani et al., "Attention Is All You Need," August 2, 2023, https://doi.org/10.48550/arXiv.1706.03762, arXiv: 1706.03762.

of what it means to relate linguistic expressions across contexts.[7] Evaluation metrics, such as Bilingual evaluation understudy (BLEU), further introduced a standardized, quantitative way of assessing output quality by comparing model-generated translations to human references.[8] While imperfect, these metrics established a framework in which performance could be iteratively optimized, a paradigm that persists in the benchmarking culture of contemporary LLM development.Similarly, pre-training strategies—initially used to improve translation quality by leveraging large amounts of monolingual data—anticipated the broader shift toward training general-purpose language models on vast corpora before fine-tuning them for specific tasks. Subword encoding methods—which in this study will later be central for our understanding of opacity—were developed to handle rare or unknown words by decomposing them into smaller units, thereby balancing vocabulary size and coverage.[9]

In this sense, modern LLMs can be understood as a generalization of machine translation research: systems that extend techniques developed for mapping between languages to the modeling of language as such. Situating machine translation historically is therefore crucial for understanding the evolving relation between language and computation: it reveals that abstractions of language give grounds to the technical decisions of how language can be encoded.

This study employs a genealogical and technically-grounded method that combines historical reconstruction with close comparative analysis of computational artifacts,

[7] Peter F. Brown et al., "The Mathematics of Statistical Machine Translation: Parameter Estimation," *Computational Linguistics* 19, no. 2 (1993): 263–311; Philipp Koehn, "Europarl: A Parallel Corpus for Statistical Machine Translation," in *Proceedings of Machine Translation Summit X: Papers* (Phuket, Thailand, September 13, 2005), 79–86, https://aclanthology.org/2005.mtsummit-papers.11/.

[8] Sourav Banerjee et al., "The Vulnerability of Language Model Benchmarks: Do They Accurately Reflect True LLM Performance?," December 2, 2024, https://doi.org/10.48550/arXiv.2412.03597, arXiv: 2412.03597; Kishore Papineni et al., "BLEU: A Method for Automatic Evaluation of Machine Translation," in *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics* (Philadelphia: Association for Computational Linguistics, 2002), 311–318, https://doi.org/10.3115/1073083.1073135.

[9] Rico Sennrich et al., "Improving Neural Machine Translation Models with Monolingual Data," in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Berlin: Association for Computational Linguistics, 2016), 86–96, https://doi.org/10.18653/v1/P16-1009; Rico Sennrich et al., "Neural Machine Translation of Rare Words with Subword Units," in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Berlin: Association for Computational Linguistics, 2016), 1715–1725, https://doi.org/10.18653/v1/P16-1162; Piotr Bojanowski et al., "Enriching Word Vectors with Subword Information," June 19, 2017, https://doi.org/10.48550/arXiv.1607.04606, arXiv: 1607.04606; Taku Kudo, "Subword Regularization: Improving Neural Network Translation Models with Multiple Subword Candidates," in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Melbourne, Australia: Association for Computational Linguistics, 2018), 66–75, https://doi.org/10.18653/v1/P18-1007.

in order to locate the specific material and historical origins of opacity in large language models. Chapter I traces the emergence of the black-box trope by reconstructing the shared technical lineage of neural machine translation and contemporary LLMs, drawing on primary architectural papers, corpus-building practices, and evaluation regimes to show how the statistical modeling of language first became productive at scale. Chapter II then isolates the precise technical substrate of this opacity by examining the input-layer transition in NMT systems, conducting a side-by-side reading of two dominant subword tokenization algorithms—Byte-Pair Encoding and Unigram LM—to demonstrate how their contrasting treatments of linguistic redundancy (greedy compression versus noise embrace) actively displace units of meaning and encode the conditions of interpretative intractability. By establishing that the reduction to textual inputs and outputs is not a methodological convenience but a historical and structural feature of the architecture itself, the analysis opens onto the epistemic consequences for media studies and STS: rather than reproducing the textual interface through familiar sociomaterial or cultural-technical lenses, the study insists on recognizing how LLMs constitutively solicit and reward textual reduction, thereby requiring scholars to treat the non-discursive substrate as constitutively illegible rather than merely opaque.

Chapter 1

METAPHORS AND HISTORIES

The 'black box' comes to scientific (and then popular) discourse from the field of cybernetics. In "Ontology of the Enemy," Peter Galison locates the term "black box" in the practical culture of the MIT Radiation Laboratory during the Second World War: the expression derived in part from the literal black-speckled metal cases used to house radar components, such as amplifiers and filters.[1] The generalization of the black box into a universal model of interaction with technology is already explicit in An Introduction to Cybernetics by W. Ross Ashby, first published in 1956. Ashby argues that black box theory extends far beyond technical laboratories: the limit example that he gives is of a child opening a door must manipulate the handle as input in order to produce the movement of the latch as output, without access to the internal linkage that connects them.[2] Everyday life, he claims, confronts us with systems whose mechanisms are not open to inspection and must therefore be approached through input–output regulation. In this formulation, opacity is not exceptional: the world is composed of systems whose interiors are functionally irrelevant for successful control; what matters is not knowledge of hidden structure but the capacity to correlate actions with effects. To adopt this framework is to imagine a universe of operational surfaces, in which all entities, not necessarily technical, can be treated as black boxes insofar as they respond to stimuli.

In such a setup, interaction replaces insight as the primary mode of study, and theory then consists in reopening what practice has rendered opaque. This shift marks the point at which the black-box metaphor ceases to be merely heuristic and becomes diagnostic of ontological assumptions. When this theoretical framework is extended, its analytic strength is found in precisely this capacity to map constraints: it differentiates between what can be reconstructed from behavior and what cannot be recovered through any manipulation of observable relations. When AI is treated as a black box in popular discourse, these are claims of ontological kind, on what AI *is* rather than how it should be treated as an object. This is the limit of the black box imagination—the world being an assemblage of isomorphic black boxes.

[1] Peter Galison, "The Ontology of the Enemy: Norbert Wiener and the Cybernetic Vision," *Critical Inquiry* 21, no. 1 (1994): 247.

[2] W. Ross Ashby, *An Introduction to Cybernetics* (London: Chapman & Hall, 1957), 86.

"Consider how many black boxes there are in the room. Open the black boxes; examine the assemblies inside. Each of the parts inside the black box is itself a black box full of parts"[3]—in Pandora's Hope, Bruno Latour uses this image to illustrate the recursive structure of socio-technical networks: every stable object is to be decomposed into further associations; each component reveals another layer of heterogeneous elements. In this formulation, what appears as a single unit is in fact an effect of prior alignments.

This recursive formulation implies a specific conception of opacity. Opacity here is always a function of tense, nested complexity. A black box is opaque because it contains further parts; opening it simply reveals additional assemblies. The same form of opacity is recurrent across different 'levels' of a technology: what is hidden at one scale, reappears as a sub-structure of another. The metaphor implies that all forms of concealment are structurally equivalent, and the black box is a product of indefinite regress of internal composition. However, once opacity is treated as a universal condition—once everything becomes a black box—the internal differentiation that made the concept useful begins to recede. If all entities are opaque in the same operational sense, then their affordances risk appearing as mere variations in scale of the same isomorphic structure. The problem is not that opacity exists everywhere, but that its particular modes may disappear under a generalized schema, leaving no criteria for distinguishing between how and why different kinds of boxes conceal their operations.

Rather than accepting the cybernetic view that all systems are black boxes in the same operational sense, the analysis in this chapter returns to the mid-twentieth-century convergence of cybernetics, information theory, and the earliest programs of machine translation. It shows how a specific reconfiguration of language—as surface input and output to an inaccessible invariant substrate—produced a form of opacity that is neither universal nor merely nested, but constitutively engineered into the architecture of statistical language processing. By tracing this lineage from wartime engineering through Shannon and Weaver's communication model to the architectural metaphors that located computational labor in a hidden "basement," the chapter demonstrates that the black-box character of LLMs is the direct, historically locatable outcome of treating natural language as separable from the mechanisms that render it computable. In this way, it lays the necessary groundwork for the

[3]Bruno Latour, *Pandora's Hope: Essays on the Reality of Science Studies* (Cambridge, MA: Harvard University Press, 2000), 185.

technical examination that follows, where the precise material site of this engineered separation will be isolated and analyzed.

1.1 Opacity Histories

Already in the mid-1930s, both Georges Artsrouni, a French-Armenian engineer, and Soviet educator and researcher Petr Petrovich Troyanskii patented designs for the first machine translation devices.[4] Both schemes were essentially mechanical dictionary systems. These efforts were characterized by the hope that electro-mechanical mechanization could speed translation through vast dictionaries or clever syntactic role encoding schemes. Hutchins emphasizes that their value was in highlighting general methods (like automated lookup and interlingual coding), even though they solved only a part of the translation problem.[5] The intellectual lineage of Artsrouni's apparatus and Troyanskii's design runs closer to lexicography—dictionary compiling—or terminological indexing than to computational semantics. Their central operation was lexical retrieval: a source word was identified, matched to an entry in a bilingual index, and mechanically reproduced or reformatted according to predetermined grammatical codes. Translation was thus conceived as the orderly selection and printing of equivalents already stored in a structured lexicon. For the purpose of the study of language-computation relation, these projects remain peripheral, because these could only be understood not as direct ancestors of contemporary machine translation systems but as mechanized extensions of the dictionary. Artsrouni and Troyanskii were pioneers in concept, but their methodologies would soon give way to digital computing approaches that develop into the field of statistical semantics. The most influential articulation of this technical possibility emerges in the work of Warren Weaver, an American pioneer of machine translation, mathematician and a science administrator. Weaver himself reflected on his role as a mediator between mathematics, communication theory, and the emerging field of machine translation,[6] positioning translation as an applied instance of broader theoretical concerns of communication. Weaver's collaboration with Claude Shannon laid the foundation for a science of information transmission, in which even non-communicative acts could be interpreted through the lens of communication

[4] W. John Hutchins, *Machine Translation: Past, Present, Future* (Chichester: Ellis Horwood / Halsted Press, 1986), 22–24; Dmitriy Y. Panov, "Machine Translation and the Human Being," *Impact of Science on Society* 10 (1960): 18.

[5] W. John Hutchins, ed., *Early Years in Machine Translation: Memoirs and Biographies of Pioneers* (Amsterdam: John Benjamins Publishing Company, 2000), 5, https://doi.org/10.1075/nlp.12.

[6] Hutchins, *Early Years in Machine Translation: Memoirs and Biographies of Pioneers*, 18.

theory. Machine translation, in this sense, is a practical instantiation of communication theory, determined by the ways in which language could be quantified and operationalized.

The Technical, the Semantic, the Effective

To situate the study of language alongside computation, it is necessary to consider the way early information theory framed communication itself. The payoff of examining the Shannon–Weaver framework at this point is to make visible the precise conceptual architecture that first rendered language computationally tractable while simultaneously installing the conditions of its opacity. By foregrounding the technical level of communication as the necessary foundation for any subsequent semantic or effective analysis, this section shows that the very possibility of probabilistic language modeling—and, by extension, of contemporary large language models—depends on a deliberate hierarchical split: symbols must first be isolated and transmitted with engineering precision before questions of meaning or social consequence can even arise. This split does not merely prioritize "superficial" engineering details; it actively produces the structural separation between observable linguistic surfaces (input and output) and an autonomous internal computational domain. In doing so, it supplies the missing link between the earlier cybernetic black-box heuristic and the ontological opacity that now defines LLMs: what appears as a neutral communication model is in fact the historical blueprint for treating natural language as peripheral interface to an inscrutable symbolic process. The analysis that follows therefore isolates this blueprint in its canonical form, preparing the ground for the architectural metaphors and technical programs that will later materialize it in machine translation.

Claude Shannon and Warren Weaver, in The Mathematical Theory of Communication, argued that the analysis of communication could be approached on multiple levels, each addressing distinct but interrelated questions. Their framework clarifies why computational approaches to language—whether in machine translation or modern probabilistic models—cannot be understood in isolation from its milieu. Weaver expresses the problems of communication in a following hierarchy under the point 1.2, Three Levels of Communications Problems:

> Relative to the broad subject of communication, there seem to be problems at three levels. Thus it seems reasonable to ask, serially:

> LEVEL A. How accurately can the symbols of communication be transmitted? (The technical problem.)
>
> LEVEL B. How precisely do the transmitted symbols convey the desired meaning? (The semantic problem.)
>
> LEVEL C. How effectively does the received meaning affect conduct in the desired way? (The effectiveness problem.)[7]

This passage establishes a hierarchy in which the manipulation of symbols (Level A) provides the necessary foundation for the subsequent questions of meaning and effect. As Weaver suggests, on first reading, Level A may appear "superficial," concerned only with engineering details, while Levels B and C carry the philosophical and interpretive weight of communication.[8] Yet this apparent superficiality is precisely what makes the computational study of language possible: accurate transmission of symbols is a prerequisite for any formal or probabilistic modeling of semantics, and ultimately, for any intervention into the social consequences of communication. Computation structures the conditions under which content (for the use of a word more precise than 'meaning') can be transferred. In other words, the technical operations of encoding, storing, and decoding language are themselves constitutive of the semantic possibilities that follow. Modern computational linguistics inherit this principle: probabilistic models of language rely on formally manipulable symbols precisely because these symbols have been rendered coherent and tractable through computational design.

The source of the sense of "superficiality" of technical operations occurs from the way a split is drawn between input and transmission. The model of communication described in The Mathematical Theory of Communication, known as the Shannon–Weaver model (Figure 1.1),[9] conceptualizes communication as a linear process in which a message is transmitted from a sender to a receiver through a defined channel. The model distinguishes key components: an information source that produces a message, a transmitter that encodes it into a signal, a channel through which the signal travels, and a receiver that decodes it back into a message for the destination. Originally formulated within information theory, the model emphasizes

[7] Claude E. Shannon and Warren Weaver, *The Mathematical Theory of Communication* (Urbana: University of Illinois Press, 1949), 4.

[8] Shannon and Weaver, *The Mathematical Theory of Communication*, 4.

[9] Shannon and Weaver, *The Mathematical Theory of Communication*, 54.

the technical problem of accurately transmitting signals rather than the semantic or interpretive dimensions of meaning.

Within this standard technical schema, what is often described as "computation" can be aligned with the transformation that occurs between encoding and decoding: the conversion of a message into a signal and its subsequent reconstruction. This transformation appears as a distinct internal operation, flanked by input (the encoding of the message by the transmitter) and output (its decoding by the receiver). This representation implies a separation between the I/O layer and internal operations of encoding and decoding: computation proper unfolds in an abstract formal space, whereas input and output belong to the contingent domain of human communication. The computational process is thus imagined as self-contained and autonomous, logically complete without regard to how its symbols are introduced or rendered.

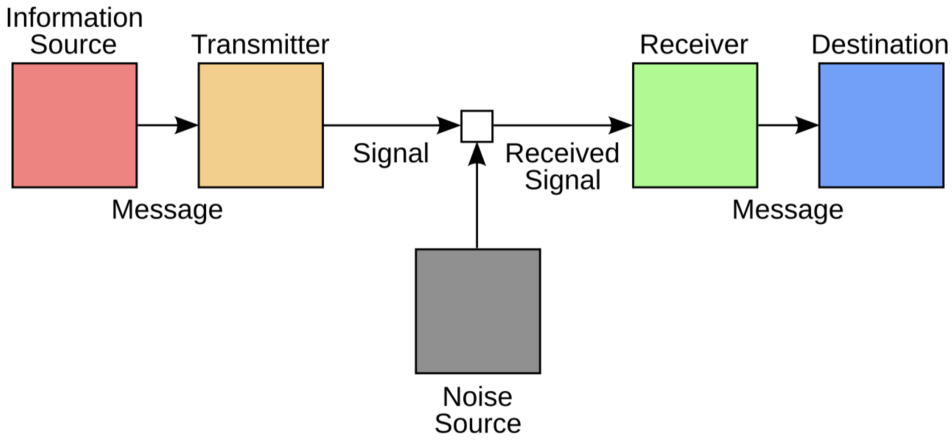


Figure 1.1: The Shannon–Weaver model of communication, adapted from the original diagram.[10]

Architectures and Architectural Metaphors

Before The Mathematical Theory Computation—in March 1947—Warren Weaver, vice president of the Rockefeller Foundation at the time, wrote to cybernetics pioneer Norbert Wiener about the translation problem.[11] Drawing on his wartime familiarity with code-breaking, he famously formulated translation as a problem of cryptography:

[10] Shannon and Weaver, *The Mathematical Theory of Communication*, 54.
[11] Hutchins, *Early Years in Machine Translation: Memoirs and Biographies of Pioneers*, 19.

"When I look at an article in Russian, I say 'This is really written in English, but it has been coded in some strange symbols. I will now proceed to decode.' [...] As a linguist and expert on computers, do you think it is worth thinking about?"[12]

In the letter, he acknowledged "multiple meanings, etc." as a semantic difficulty but hoped a machine could at least manage scientific text.[13]

Wiener's April 30, 1947 reply was cautiously pessimistic. He thanked Weaver and noted that simple languages (like Basic English) made some things easier, but warned that natural language is not like code. As he put it, the "boundaries of words in different languages are too vague and the emotional and international connotations are too extensive to make any quasi-mechanical translation scheme very hopeful".[14] Wiener admitted Basic English could in theory help, but pointed out that even its simple grammar (e.g. the verb "get") required heavy semantic burden. He concluded that beyond trivial applications, such as reading aids for the blind, full mechanization of language seemed "very premature".[15] In the last response, Weaver still believed that computation could brute-force much of the ambiguity by sheer volume of case. This exchange highlights the conceptual tension of the era; Weaver's view, influenced by cryptography and Shannon's nascent information theory, was optimistic: he treated language as codable and believed statistical methods might overcome ambiguity. Wiener's view was skeptical: he saw words and meanings as too fluid and subtle for simple mechanization. In their correspondence Weaver framed translation as a kind of code-breaking, and Wiener reminded him that unlike ciphers, natural languages carry heavy nuance.

Nevertheless, Weaver pressed on. Within two years he circulated his famous "Translation" memorandum dated July 1949, drawing on ideas from these discussions, wartime code-breaking successes, and new information theory. A related but distinct articulation of translation's language environment appears in the memorandum, where he reflects on wartime cryptography as a model for translation. He writes:

> During the war, when the whole field of cryptography was so secret, it did not seem discreet to inquire concerning details of this story; but one could hardly avoid guessing that this process made use of

[12] Warren Weaver, "Translation," in *Proceedings of the Conference on Mechanical Translation* (1952), 52, https://aclanthology.org/1952.earlymt-1.1/.

[13] Weaver, "Translation."

[14] Weaver, "Translation."

[15] Weaver, "Translation."

> frequencies of letters, letter combinations, intervals between letters and letter combinations, letter patterns, etc., which are to some significant degree independent of the language used[16]

Here, the probabilistic imagination of language acquires an explicitly interlingual dimension. Whereas the discussion in The Mathematical Theory of Communication demonstrates that English can be described through statistical regularities internal to the language, the memorandum extends this logic across languages. The crucial claim is that frequencies and patterns are "to some significant degree independent of the language used": translation thus becomes conceivable not because languages share meanings in any straightforward philosophical sense, but because they may share structural regularities at the level of distribution. Cryptography serves as the mediating analogy: just as encoded messages can be deciphered through statistical analysis without prior knowledge of their content, so too might one approach translation by exploiting cross-linguistic regularities in symbol patterns. Warren Weaver uses the term "language invariant" to refer to structural or statistical features of language that remain stable across different natural languages and can therefore serve as a basis for translation. He hypothesizes that there exist regularities—such as frequencies of letters, word patterns, syntactic arrangements—that are sufficiently independent of any particular language to permit formal comparison. These invariants would function analogously to stable features exploited in cryptanalysis: underlying patterns that persist beneath surface differences.

In the final pages of his 1949 memorandum, Warren Weaver proposes a path forward for mechanical translation by invoking an architectural metaphor.[17] He asks readers to picture speakers of different languages as individuals in tall, closed towers, each shouting toward the others; communication is difficult because messages must travel horizontally from one isolated structure to the next. The solution, according to Weaver, occurs when people descend from their towers into the open basement that lies beneath all of them—a shared foundation where exchange becomes possible. Applied to translation, this means that moving directly from Chinese to Arabic or from Russian to Portuguese is inefficient; instead, one should descend from each language to the underlying common base—an as-yet-undiscovered universal logical structure—and then rise back up through a convenient route into the target language. Weaver envisions a level of analysis so deep that it reveals structural regularities

[16] Weaver, "Translation."
[17] Weaver, "Translation."

amenable to formal, algorithmic treatment—precisely the kind of invariants that could be encoded and processed by a machine. Only after this descent into the shared computational substrate does re-ascent into any specific language become feasible.

Precisely this metaphor creates a conceptual split between I/O layer and operations: by locating the intellectual and technical labor in the invariant basement—the work in the logical structure of languages—Weaver assigns the towers themselves, that is, the particular grammars, vocabularies, and surface forms of real languages, to the role of entry and exit points. The input layer (the source text) and the output layer (the translated text) become portals that deliver raw material downward and release the finished product upward. In this way, the imagery through which early machine translation is theorized establishes a divide between the technical operations in the common foundation and the peripheral input/output layers that frame the process.

1.2 Anti-Babel

The black box of AI is not an accident of complexity, nor a universal feature of all technology, nor a mythical aura. It is the direct consequence of a historical decision to treat the surface of language as separable from a computational invariant layer. Once the machine's labor was located in the inaccessible basement of parameters, the very architecture (with the metaphors of tower, the very word obtaining a rather literal meaning) renders structure irrelevant to the observable relations between input and output. In this sense, the persistent rhetorical emphasis on the black box is not redundant; it names the singular historical achievement—and the singular limitation—of a technology whose power derives precisely from the depth at which it has rendered language operational while rendering its own operations inscrutable. AI is a black box because the path between the input and output is arbitrary as long as it is convenient. One might object that the tower-and-basement metaphor is merely a rhetorical image Weaver used in 1949, and that such metaphors do not shape actual engineering decisions. Yet this objection underestimates the real influence of architectural imaginaries in the history of computing. Weaver's figure was not a casual illustration. It crystallized a foundational conceptual split: between the surface forms of natural languages (the isolated "towers") and a deeper, language-invariant computational substrate (the shared "basement") where the real work of translation was imagined to take place. Far from remaining a mere metaphor, this image and the communication-theoretical framework it embodied actively shaped the research program that followed: it licensed the prioritization of statistical regularities and

probabilistic modeling over explicit semantic or grammatical rules. What began as a metaphorical descent into a common foundation became materially inscribed in the design of systems that treat linguistic input and output as peripheral interfaces to an opaque internal process. In the following chapter, I will zoom into the technical moment of transition from the observable surface of linguistic input to demonstrate in what particular ways computation is constitutively defined by it. Far from being a peripheral interface added to an autonomous formal core, language supplies the very medium in which symbols become tractable, regularities become computable, and outputs become intelligible; it is the condition under which the machine can be said to "compute" at all. Yet precisely in the course of this transition the possibility of direct access to the operations themselves is lost, structurally.

Chapter 2

THE UNIT OF LANGUAGE

The question of defining and separating the units of language is of no concern for a speaker of a language: whatever these units are, their interplay can exist without delineation. But to operationalize meaning transmission, one must determine meaningful units: an alphabet and the meaning-bearing elements composed of these alphabetic symbols. In this sense, the process of machine translation doubles: to mechanically translate from one natural language into another, one must first translate from natural language to the language of computation. In this section, I am going to zoom into this process of translation between the natural language of the input and the calculative language of operations in order to establish what we overlook in treating these as separate and independent. In case of statistical and neural machine translation, this process occurs in the tokenization algorithms. By examining how tokenization actually works, we can see that the apparent separation between natural language and computation is the precise site where the conditions for later black-box opacity are first materially inscribed, and the way in which it is done is not singular. Far from being a minor preprocessing step, tokenization quietly determines what counts as a "unit" of language inside the model, thereby shaping both the limits of interpretability and the very possibility of statistical language modeling.

Tokenization can be defined as the process of segmenting a continuous stream of text into discrete, atomic units—termed tokens—that serve as the fundamental input to computational models.[1] Critically, this process marks the fundamental transition from a linguistic object to a computable form. Raw text—a variable string of characters—is transformed into fixed numerical sequences that can be processed mathematically. Tokenization thus functions as the essential interface where human language is first rendered legible to machines. In this way, it is exactly tokenization that serves as the essential interface where human language is rendered legible to machines. To understand the role of tokenization in LLMs, we must identify what problems coming from the field of translation have determined the trajectory of tokenization development. This chapter is a story of how the unit of meaning has

[1] Sabrina J. Mielke et al., "Between Words and Characters: A Brief History of Open-Vocabulary Modeling and Tokenization in NLP," December 20, 2021, https://doi.org/10.48550/arXiv.2112.10508, arXiv: 2112.10508.

been transformed by neural machine translation technologies—becoming inherently non-linguistic—and how the operations of NMT have been obscured in this process of redefinition.

Tokenization does more than simply prepare text for a model — it directly confronts and operationalizes two of the most persistent difficulties in machine translation. The first difficulty is the open vocabulary problem: natural languages constantly generate new words, compounds, names, and slang that no fixed dictionary can ever fully anticipate. The second difficulty is structural misalignment between languages: what counts as a meaningful unit in one language often does not line up neatly with units in another. Words may be split differently, meaning may be carried by prefixes or suffixes in one language but by separate words in another, and grammatical structures vary widely.

Tokenization addresses both challenges at their root. By breaking raw text into smaller, reusable subword units (rather than trying to treat whole words as indivisible atoms), it creates a flexible, finite vocabulary that can handle unseen or rare forms through recombination. At the same time, these subword units make it possible to align linguistic material across languages more effectively, turning the messy realities of linguistic difference into manageable patterns that a model can learn statistically. In short, tokenization quietly converts the classic, seemingly intractable problems of machine translation into problems of representation that the neural network can actually work with. When these distinct media operations are blurred under a generic "subword" label, we risk obscuring how different inscription strategies shape the model's capacity for compositional insight.

Tokenization is the quiet but decisive first inscription through which the multiplicity of natural language is pressed into the discrete symbolic order of neural machine translation systems. Before any attention mechanism fires or any semantic vector is computed, tokenization has already performed a foundational mediation: it breaks the utterance into probabilistic units that are only accessible through algorithmic logics and in no way are accessible through the logics of spoken language.

This initial step shifts the entire subsequent computational process outside the register of the linguistic unit of meaning and into a regime governed by statistical compression and probabilistic segmentation. The task of this chapter, then, is neither to stage an oppositional comparison between tokenization methods nor to offer an exhaustive technical description of their mechanics. Its purpose is twofold: first, to demonstrate that it is tokenization itself that first forces operations

of NMT (and LLMs by extension) outside the register of the linguistic unit of meaning; and second, to show how this displacement simultaneously obscures the particularity of different tokenization algorithms. This particularity is available to us in tracing how the problems of redundancy is treated across different solutions. I am looking at Byte-Pair Encoding (BPE) and Unigram algorithms to illustrate two relations: BPE's fighting of linguistic redundancy through deterministic merges, and Unigram's controlled noise injection and recognition of numerous plausible segmentations. These present us the fact that units of meaning in natural language generation are not just non-linguistic, but are particular in the ways they decline the linguistic register.

Identifying the Unit of Meaning To compute natural language, it must first be converted from ordinary human text into a numerical form that machines can process through arithmetic operations. In a way, the fact of conversion of language into manipulable units was already well-understood before the emergence of computational linguistics. How does one identify such a unit?

2.1 Linguistic Units

In Course in General Linguistics, Ferdinand de Saussure proposes that language can be provisionally represented as two parallel chains: one composed of concepts and the other of sound-images (chains 'A' and 'B' respectively in Figure 2.1). These chains are not independently structured and then aligned; rather, they are articulated only through their mutual correspondence. What appears as a "unit" of language is not given in advance within either chain, but emerges at the point where a segment of sound is associated with a segment of meaning. The schema thus serves to demonstrate that segmentation in language is not a purely physiological or purely conceptual operation. A continuous stream of sound cannot be divided into meaningful parts by phonetic criteria alone, just as a flow of thought does not come pre-partitioned into discrete concepts. Instead, divisions become linguistically relevant only when a cut in the sound chain coincides with a cut in the conceptual chain.

Saussure further clarifies this point by arguing that isolating either side of the sign dissolves the linguistic unit. A succession of sounds becomes linguistic only when it supports an idea; otherwise it is simply material for phonetic or physiological analysis. Conversely, psychological concepts such as "house," "white," or "see" become

linguistic only when attached to a sound-image.[2] Saussure therefore concludes that "a concept is a quality of its phonic substance just as a particular slice of sound is a quality of the concept".[3] To illustrate the inseparability of these components he compares the linguistic sign to a chemical compound: "a better choice would be a chemical compound like water, a combination of hydrogen and oxygen; taken separately, neither element has any of the properties of water".[4] The unit of language is neither sound nor meaning alone but their structural union within the system of signs.

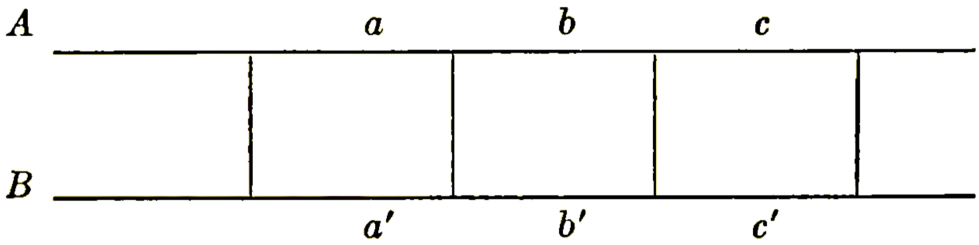


Figure 2.1: Ferdinand de Saussure's delineation of linguistic units.[5]

This definition is what allows Saussure to map units of language as language is represented in writing. Just as the temporal chain of sounds is segmented only where phonic cuts coincide with conceptual cuts, the spatial chain of graphic marks in writing makes those same units immediately visible: spaces between words, for instance, correspond precisely to the points of articulation established by the system of signs. Writing is a form of naive partitioning, where the otherwise continuous stream of speech appears already partitioned into discrete entities—each a stable union of a graphic image and a concept. The schema therefore functions equally for both spoken and written language, demonstrating that the linguistic unit is always defined internally by the langue rather than by the material properties of its medium. Writing renders units' systemic delimitation apparent; these delimitations are not stable, but at the end these could only be sentences, idioms, words, and morphemes, ultimately composed of characters of the alphabet. These determine the naive

[2]Ferdinand de Saussure, *Course in General Linguistics*, ed. Charles Bally and Albert Sechehaye, trans. Wade Baskin (New York: Philosophical Library, 1959), 103.

[3]Saussure, *Course in General Linguistics*, 104.

[4]Saussure, *Course in General Linguistics*, 103.

[5]Saussure, *Course in General Linguistics*, 104.

toolbox of language operationalization: conveniently, many written languages are split into words and sentences by spaces and punctuation.

Ideally, these would give ground to the automatic splitting of utterances into units; this is not the case. It is Norbert Wiener who was the first one to recognize the complexity of identifying the operational unit of meaning when it comes to machine translation. In his response to Weaver's letter mentioned in the previous chapter, he gives his pessimistic caution:

> ...as to the problem of mechanical translation, I frankly am afraid the boundaries of words in different languages are too vague and the emotional and international connotations are too extensive to make any quasi-mechanical translation scheme very hopeful.[6]

At the time, even the fantasy of a context-sensitive translation machine was impossible, for that the issue of unit delineation already is hard and is amplified by the plurality of language structures. Later, Yehoshua Bar-Hillel, the author of the first systematic critique of early machine translation experiments (first published in 1953), points out the insufficiency of our unit-determination strategies:

> Somehow one can envisage how a machine could proceed in a kind of word-by-word translation but it is exactly this type of translation which collapses when confronted with an idiom which by definition, to wit, definition 3 of Webster's Collegiate Dictionary. 1951, is "an expression in the usage of a language, that is peculiar to itself either in grammatical construction or in having a meaning which cannot be derived as a whole from the conjoined meanings of its elements (as, the more the merrier, a picture of the king's, to make friends with him)."
>
> Any structural linguist would immediately be put on his guard by the fact that in this definition the word "meaning" occurs, and even twice. What exactly is meant by "conjoined meanings of its elements"? How does one determine that, to utilize one of the examples mentioned in Webster, the meaning of the phrase "to make friends with him" is not derivable as a whole from the conjoined meanings of the elements of

[6] Weaver, "Translation."

> this phrase? And what are these elements? Words? Morphemes? And where does the order of the elements come into the picture?[7]

Bar-Hillel's critique, building directly on Wiener's earlier warning about the "vague boundaries of words" and the intractable "emotional and international connotations," exposes the precise fracture line that Saussure's schema had already implied but that early machine-translation designers had ignored. When a machine is asked to translate, it must first decide where to cut the foreign chain of symbols and where to re-anchor those cuts in the target conceptual chain; yet the very notion of a stable "element" (word, morpheme, phrase) dissolves the moment one confronts an idiom whose meaning is not the arithmetic sum of its parts. The double appeal to "meaning" in Webster's definition is yet another symptom: without an internal system of differences that defines what counts as a unit in each language, there is no principled way to determine which segments are "conjoined" or why their order matters. The operational unit cannot be read off the surface of writing or speech, taken as granted in the form of words; it appears that it must be reconstructed anew inside every written language.

If not for the word, there is a stable element that seems to occur across most written languages, and it is the letter. Genealogies of machine translation that extend beyond twentieth-century computing reveal that the delineation of operational units at the level of the character precedes modern linguistic formalization and is rooted in cryptanalytic practice. As DuPont argues in his article "The Cryptological Origins of Machine Translation, From al-Kindi to Weaver," this kind of thinking can be traced back to medieval Arabic cryptanalysis, where letters constituted the primary units of analysis subject to counting, comparison, and recombination.[8] Techniques such as letter-frequency analysis, vowel–consonant ratios operationalized text as statistically tractable sequences rather than semantically coherent wholes, thereby establishing a mode of processing indifferent to higher-order linguistic units like words or syntax. Alternative genealogy of character-based analysis would follow from the work of Andrey Andreevich Markov the Elder, who analyzed literary text—most notably a 20,000-character long excerpt from Alexander Pushkin's Eugene Onegin in 1913—by converting it into a sequence of discrete symbols and

[7] Yehoshua Bar-Hillel, "Some Linguistic Problems Connected with Machine Translation," *Philosophy of Science* 20, no. 3 (1953): 220–21, https://doi.org/10.1086/287266.

[8] Quinn Dupont, "The Cryptological Origins of Machine Translation, from al-Kindi to Weaver," *Amodern* 8 (January 2018).

studying the statistical dependencies between successive elements.[9]

This whole chapter could have been a media history of machine translation, and not a media theory of it, if character-based encoding would be viable in the material conditions of machine translation; it is not—for the factor of computational cost. Character-based tokenization, despite its theoretical appeal in overcoming limitations of subword methods such as poor handling of morphological richness and vulnerability to source-side noise, remains a suboptimal choice for neural machine translation in practice.[10] Character-level systems exhibit no clear advantages in domain robustness or morphological generalization—the very motivations often cited in the literature—while suffering from substantially higher computational costs due to longer sequences, making them impractical for competitive setups.[11] The only notable benefit is greater robustness to synthetic source noise, but this does not outweigh the drawbacks in translation fidelity and training/inference efficiency.[12] The task of tokenization becomes finding an optimal non-linguistic unit of meaning.

At this point, we are able to identify the central drama of natural language generation: no meaning-bearing unit that we can identify and recognize as speakers (and writers)—sentences, words, letters—suffice in making language operable. The advancements of natural language generation are only possible in the negation of these units and the overwriting that occurs in the process of tokenization. This is the point at which the obscuring of technical operations occurs, pushing those outside the register of any languages we use. For now, we have seen that the units of everyday language—sentences, words, individual letters—do not provide a stable foundation for computation. From Wiener's pessimism and Bar-Hillel's critique to the historical reliance on character-level analysis in cryptanalysis and early statistical methods, the section has shown that no pre-existing linguistic unit can be directly operationalized without fundamental difficulties in handling open vocabularies, idiomatic expressions, and cross-linguistic misalignment. Then, the real drama of natural language generation lies precisely in the moment of tokenization. Here the model must actively negate and overwrite the familiar units of human language,

[9] Andrey A. Markov, "An Example of Statistical Investigation of the Text Eugene Onegin Concerning the Connection of Samples in Chains," *Science in Context* 19, no. 4 (2006): 591–600, https://doi.org/10.1017/S0269889706001074.

[10] Jindřich Libovický et al., "Why Don't People Use Character-Level Machine Translation?," in *Findings of the Association for Computational Linguistics: ACL 2022* (Dublin, Ireland: Association for Computational Linguistics, 2022), 2470–2485, https://doi.org/10.18653/v1/2022.findings-acl.194.

[11] Libovický et al., "Why Don't People Use Character-Level Machine Translation?," 2477.

[12] Libovický et al., "Why Don't People Use Character-Level Machine Translation?," 2471.

replacing them with its own statistical subword units. It is at this exact point that the constitutive obscuring of technical operations occurs: language is made computable only by being translated into a non-linguistic numerical form whose internal logic is no longer accessible or describable in the registers of ordinary language.

2.2 Computational Units

We have by now established that the two naive ways of delineating units—over words and over characters—are both problematic, though they did give the start to the development of tokenization algorithms. As Mielke et al. observe, "In teaching settings and in fact historically in NLP, these tokens are somewhat naturally implied to be words—at first, perhaps naively as 'space-separated substrings' in English".[13] First forms of tokenization are based on a foundational intuition that the most immediate and accessible unit of text is the word, demarcated conveniently by whitespace. Yet this apparent simplicity, the authors suggest, is deceptive: the moment one encounters phenomena such as punctuation attachment or contractions—authors suggest the word "don't" as one example[14]—the clean boundary implied by spaces begins to dissolve, revealing that even this ostensibly basic segmentation choice rests on assumptions that do not hold uniformly across language use.

In designing an optimal variable-length code for a message, the principle of assigning shorter codewords to more probable messages is cenral—as described by the American computer science pioneer David W. Huffman, a graduate student in Robert Fano's information theory course at MIT. Fano, who had co-developed the Shannon–Fano coding method with Claude Shannon, assigned his students a problem: find the optimal binary code—the one that minimized average code length for a given set of symbols.[15] Huffman, in responding to this problem, has formalized the underlying principle of every modern tokenizer (and truly, one of the rules of human-machine communication in general)—that the more frequent units must have more compact representations.

Consider a simple communication system with three messages: "Hello," used with high frequency; "I'm hungry," used occasionally; and "There's a fire!," used rarely. If the shortest codeword were allocated to "I'm hungry" rather than to "Hello," the

[13] Mielke et al., "Between Words and Characters: A Brief History of Open-Vocabulary Modeling and Tokenization in NLP," 1.

[14] Mielke et al., "Between Words and Characters: A Brief History of Open-Vocabulary Modeling and Tokenization in NLP," 2.

[15] Gary Stix, "Profile: David A. Huffman. Encoding the "Neatness" of Ones and Zeroes," *Scientific American* 265, no. 3 (1991): 54.

system would be inefficient, because the frequently used message would require a longer representation. Simply swapping the codewords so that "Hello" receives the shortest code would reduce the average codeword length, thereby improving overall efficiency. This reasoning underlies the fundamental requirement that, in an optimal code, no less probable message may receive a shorter codeword than a more probable one.

Furthermore, when two messages share identical probabilities, their codeword lengths may differ without affecting the average code length, as any permutation of their assignments yields the same expected performance.[16] Consequently, for the purpose of code construction, we can assume the messages are ranked from most to least probable, ensuring that the shortest codewords are systematically assigned to the most frequent messages. This principle of optimal assignment, formalized by Huffman, stands in direct tension with the role of redundancy in communication systems—Huffman's algorithm embodies the pure logic of compression, pushing a source toward its entropy limit, whereas natural languages and human communication deliberately tolerate high levels of redundancy precisely because that surplus enables reliable decoding even when parts of the signal are corrupted or lost.

In communication theory, redundancy designates the portion of a message that exceeds the minimum information required for its transmission, formalized within the framework established by Claude Shannon. More precisely, redundancy can be defined as the difference between the total encoded signal and the actual informational content measured in terms of entropy, such that it represents a surplus of structure or repetition within the message.[17] Although this surplus may appear inefficient from the standpoint of optimal encoding, it performs a crucial functional role: redundancy counteracts the effects of noise in a communication channel by enabling interpretive recovery.

Shannon poses redundancy as an operational resource. When he observes that "no attempt is made to eliminate"[18] the redundancy of English in ordinary transmission (as in our day to day communication), the point is that natural language is not optimized for maximal compression but for resilience. In principle, the source could be encoded more efficiently—reducing transmission time by eliminating predictable

[16] David A. Huffman, "A Method for the Construction of Minimum-Redundancy Codes," *Proceedings of the IRE* 40, no. 9 (1952): 1099, https://doi.org/10.1109/JRPROC.1952.273898.

[17] Shannon and Weaver, *The Mathematical Theory of Communication*, 56.

[18] Shannon and Weaver, *The Mathematical Theory of Communication*, 75.

structure—but this would also remove the very patterns that enable reconstruction under noisy conditions. The famous approximation of English being 50

Redundancy in translation becomes evident when a source message can undergo partial loss or alteration without preventing the recovery of its meaning in the target language. Because natural languages encode information across multiple overlapping levels, a translator is never truly dependent on a one-to-one correspondence between individual units: if a sentence contains ambiguous or untranslatable elements, the surrounding context often constrains interpretation sufficiently to allow an adequate rendering. Idiomatic expressions illustrate this particularly well: even when their literal components do not map directly onto the target language, their communicative function can be reconstructed through contextual redundancy. In this sense, translation does not operate on isolated units but on a surplus of signals that in sum stabilize meaning. Redundancy thus enables translation to tolerate indeterminacy and partial mismatch, as the loss of specific elements can be compensated by the persistence of structure elsewhere in the message.

2.3 BPE and Unigram

I want to introduce the particular ways in which tokenization algorithms operate exactly in their relation to the described redundancy. This following section is not about which method is better, which crystallization of this relation is the most productive, operationally; this is not a story of evolution in transition to the more and more advanced algorithms. This is an account of the particular strategies used in determining the counterintuitive, inhuman unit in which meaning is transmitted. Two sub-word segmentation methods that I am looking at are coming from different genealogies and encode different relations to the problem of redundancy.

Byte Pair Encoding (BPE) originates in data compression, where redundancy is iteratively reduced by greedily merging the most frequent adjacent symbol pairs, turning recurring sequences into single tokens. In contrast, the Unigram model arises from statistical machine translation, where redundancy is modeled probabilistically by assigning likelihoods to an overcomplete set of subword units and selecting the most probable segmentation. Thus, BPE operationalizes redundancy as compressible repetition fixed into the vocabulary, while Unigram treats it as distributional uncertainty across competing segmentations. These distinct genealogies yield different "inhuman" units: one crystallized through deterministic compression, the other through probabilistic inference.

Byte Pair Encoding (BPE), originally introduced in the domain of data compression by Gage in 1994 and later adapted for NLP by Sennrich et al., operates on a greedy, iterative merging process.[19] The algorithm begins by treating each character (or byte) in the training corpus as an individual token. It then iteratively identifies the most frequent pair of adjacent tokens and merges them into a new token, adding this merged unit to the vocabulary. This process continues until a predefined vocabulary size is reached or no further beneficial merges remain. The core intuition behind BPE is frequency-driven compositionality: frequently co-occurring character sequences—common prefixes, suffixes, morphemes—become single tokens, enabling the model to represent rare or unseen words through combinations of learned subwords.

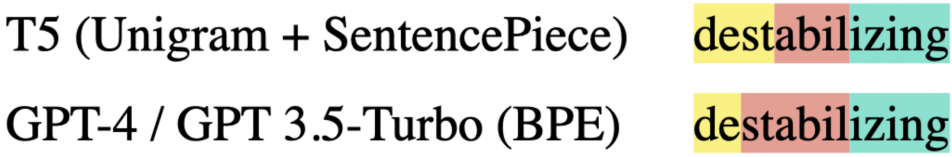


Figure 2.2: An illustration of Unigram and BPE subword tokenization.

The training of BPE is deterministic and bottom-up. It relies solely on pair-wise frequency counts without probabilistic modeling. Once the merge rules are learned from the corpus, tokenization of new text follows these rules in the exact order they were acquired, applying merges greedily from left to right; this results in a fixed set of merge operations. BPE's strength lies in its simplicity and efficiency: it produces a vocabulary that balances coverage and compactness, leading to fewer unknown tokens compared to pure word-level tokenization. Moreover, because it builds longer tokens from frequent pairs, it naturally captures morphological regularities, such as English plural forms ("cat" + "s" $\rightarrow$ "cats") or compound words.

In contrast, the Unigram Language Model approach to tokenization, proposed by Kudo in 2018 as part of the SentencePiece framework, adopts a probabilistic, top-down methodology grounded in statistical language modeling. Rather than starting

[19] Philip Gage, "A New Algorithm for Data Compression," *C Users Journal* 12, no. 2 (1994): 23–38; Rico Sennrich et al., "Neural Machine Translation of Rare Words with Subword Units," in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Berlin: Association for Computational Linguistics, 2016), 1715–1725, https://doi.org/10.18653/v1/P16-1162.

from characters and merging upward, Unigram begins with a large superset of potential subword candidates—often generated by exhaustively extracting all possible substrings up to a certain length from the training data or using heuristics like BPE-initialized candidates. It then trains a unigram language model over this candidate set, assigning a probability to each subword based on its occurrence likelihood in the corpus under the assumption that tokens are independently generated. Pruning occurs iteratively: at each step, the algorithm evaluates the loss in likelihood when removing a subword and eliminates those that contribute the least to the overall probability.

Figure 4. Subword sequences encoding the same sentence "Hello World." Taku Kudo, "Subword Regularization: Improving Neural Network Translation Models with Multiple Subword Candidates," Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics, 2018.

Before any attention mechanism activates, the continuous stream of natural language has already been pressed into a regime governed by statistical compression and probabilistic segmentation. What enters the model is a sequence of subword atoms whose boundaries are dictated by algorithmic attempts to solve these very translation difficulties. BPE addresses the open-vocabulary problem through aggressive, deterministic compression, fighting redundancy by collapsing frequent patterns into fixed tokens that allow rare or unseen forms to be recomposed from learned fragments; yet this same greedy logic can fracture morphological coherence and obscure productive prefixes when languages distribute meaning differently. Unigram, meanwhile, confronts the same problems by embracing multiplicity: it explicitly acknowledges that any word admits numerous plausible segmentations and uses controlled noise injection to turn redundancy into a generative resource, producing units that are more resilient to structural misalignment across languages. In this way, the old anxieties articulated by Wiener and Bar-Hillel—the vagueness of word boundaries and the non-additive nature of idiomatic meaning—are relocated and operationalized at the sub-word level. It is here that neural machine translation (and large language models more broadly) are first forced outside the register of the linguistic unit of meaning and into a calculative order whose primary concern is enumerability.

Yet because these distinct inscriptional regimes are routinely gathered under the generic banner of "subword tokenization," their particularity—and the specific ways they reshape translation problems—is easily obscured. When BPE's fight against

redundancy and Unigram's cultivation of noise are treated as interchangeable technical variants, we lose sight of how each produces a different kind of legibility for the model.

This chapter has traced the journey of the linguistic unit—from Saussure's langue, where meaning and sound are inseparable as two sides of a chemical compound, to its dissolution and reconstruction within the calculative order of neural machine translation. We have seen that the units which ground our everyday linguistic self-understanding—words, morphemes, even letters—prove inadequate for the operational demands of computation, falling victim either to the "vague boundaries" Wiener identified or to the prohibitive computational costs of character-level processing. The resulting units, forged through tokenization algorithms like Byte-Pair Encoding and Unigram, rewrite the conditions of meaning transmission. This rewriting, however, comes at the cost of obscuring the distinct inscriptional strategies that produce it. Tokenization occurs as a penetrable membrane, in the transition through which redefinition of a meaningful unit occurs—simultaneously with the erasure of legibility of technology's operation.

Chapter 3

FORMS OF REDUCTION

In his book Embodying Technesis: Technology Beyond Writing, Mark B. Hansen diagnoses a persistent tendency in poststructuralist and cultural theory to reduce technology to discourse in two acts of reduction—the reduction of technology to machine and the reduction of machine to text. Hansen coins the term technesis, the putting-into-discourse of technology. This double reduction highlights how theory often approaches technology only through familiar conceptual categories or metaphors—in Hansen's analysis these are examples of poststructuralist thought: "Heidegger's horizon of thought, Derrida's closure of representation, Freud's psychic space, Lacan's objet a, and Deleuze and Guattari's social assemblage"[1]—which may obscure the material and affective dimensions of technology. Finding such a reduction in contemporary theory of technology puts at question the relevance of the very theory: it allows us to ask what is actually the object of the study of technology?—implying that the object that we have been talking of has never been technology itself; reduction is theory's pharmakon: it allows us to talk technology, but deprives us of technology's particularity.

As we have seen in the Chapter I, historically, and in Chapter II, operationally, natural language generation technologies force us outside their technical operations by redefining the meaningful unit of language as sub-word, non-linguistic. If technesis is not only a procedural habit of analysis but an affordance shaped by the object itself, then the epistemic costs traditionally attributed to methodological choices must be reattributed, at least in part, to the technologies under examination. Analyses that continue to treat LLMs through the familiar lenses of discourse—cultural techniques, sociomaterial assemblages—risk participating in the very reduction the systems already perform at an architectural level. By examining how large language models instantiate and radicalize technesis, this chapter argues that the reduction of technology to text and discourse is no longer primarily a methodological limitation of media studies or STS. Instead, it emerges as a productive affordance of the technology itself. LLMs are engineered such that their operational core—statistical regularities distributed across subword token sequences—remains constitutively

[1]Mark B. Hansen, *Embodying Technesis: Technology beyond Writing* (Ann Arbor: University of Michigan Press, 2000), 20.

inaccessible to linguistic or symbolic analysis, while the system delivers fluent, human-like text at the surface. What these fields have long enacted discursively through procedures of stabilization (whether via cultural techniques, rules of mediation, sociomaterial negotiations, etc.) is here materialized architecturally: the model's power and apparent intelligibility derive precisely from sustaining a textual interface that systematically obscures its own non-linguistic operations.

This reversal reframes the epistemic stakes for media studies and STS. Rather than critiquing our own disciplinary habits of reduction, we must now attend to how certain technologies actively invite and reward text-centered modes of analysis. The black box of contemporary AI, in this light, is not a generic opacity awaiting sociological or discursive unpacking, but a historically achieved condition in which language is rendered computable only by detaching its surface performance from the sub-linguistic machinery that produces it. Recognizing technesis as an affordance rather than a mere procedural necessity compels us to ask what dimensions of technology—and of language itself—become illegible when the object under study is designed to perform its reductions for us.

3.1 Understanding Reduction

In Chapter 3 of Embodying Technesis, Hansen critiques the reduction of technology to the metaphor of the machine through a close examination of examples tracing back to the 19th century and developing throughout the 20th. He begins by analyzing two dominant critical modes: representationalist discourse, exemplified by Michelle Kendrick, who acknowledges the urgency of approaching technology as beyond user control yet still filters its effects through the representational construction of the self—in which exteriority is only possible as relational exteriority internal to "the representational space of thought";[2] and systems-theoretical approaches like William Paulson's, which frame literary texts through cybernetics and complexity theory presupposing the homology of literary and technoscientific knowledge. Hansen argues that both frameworks reduce technology to a relative exteriority internal to thought, thereby insulating it from any disruptive material force. The historical roots of this tendency towards metaphor date back to modernist and symbolist literature, on which Hansen cites Wylie Sypher's reading of Mallarmé's Un Coup de Dés as a technicist attempt to eliminate chance through formal control. The peak manifestation of this reduction is found in a metaphor of the poem as a machine—found in Paul Valéry and William Carlos Williams—meaning mechanistic,

[2]Hansen, *Embodying Technesis: Technology beyond Writing*, 78.

freed from "the casual, the contingent, the irrelevant".[3] In all these instances, the metaphor of text—machine repeatedly projects a particular reductive notion of the machine as a closed, formal system—rule-bound, devoid of contingency.

Each of these examples serves Hansen's broader argument: it is only the identification of technology with the machine metaphor that allows critical thought to maintain control over the radical material alterity of technology. Metaphors do not simply illustrate pre-existing concepts; rather, they organize the field of intelligibility within which concepts emerge. In this case, the hidden doubling of the representational metaphor—technology as a machine, and then machine as a text—is what falsely structures our ideas about technology as fully accessible to language as we understand it, and at the time as closed and stable like mechanisms.

The terms that Hansen suggests to analyse the history of philosophy of technology can be extended productively to contemporary research on technology. Technology is a common object of inquiry across several disciplines coming from the humanities and social sciences—in our case, media studies and STS. While these disciplines rightly frame their engagement with technology differently, they nonetheless proceed from the shared, often unexamined assumption that they are all examining one and the same entity. This tacit consensus is more fragile than it appears: "technology" functions here less as a clearly delimited referent than as a floating signifier whose coherence is retroactively secured by each discipline's own analytical operations. The methodological distinction between these disciplines therefore lies not primarily in their choice of objects or empirical materials, but in their differing answers to a more foundational question: how to pose technology discursively. I refer here to the basic analytical operations by which technology is stabilized as an object of inquiry—for example, as the realization of a technique, as an assemblage, as a historical process, as a sociomaterial practice, as an actor, as a vehicle for power relations. Each of these framings is performative: it does not simply describe technology but actively constitutes it as a coherent entity suitable for scholarly attention.

This rendering is not an optional interpretive step; it is a necessary condition for the very possibility of inquiry. The object "technology" is constitutively unstable—its boundaries shift depending on whether one emphasizes tools, systems, practices, affects, infrastructures, discourses—and every discipline must therefore perform a preliminary stabilization to define what technology is within its own epistemic

[3] Hansen, *Embodying Technesis: Technology beyond Writing*, 82.

frame. These fields readily acknowledge this instability, yet each grasps it differently: some treat it as an ontological feature to be thematized, others as a methodological problem to be managed, and still others as a productive ambiguity that invites perpetual deconstruction. Crucially, each stabilization procedure selects in advance the legitimate forms of evidence and the appropriate scale of analysis. In doing so, it simultaneously excludes or renders invisible those dimensions of the technological that cannot be captured or articulated within the chosen discursive framework. What counts as relevant data, what temporal or spatial scale is deemed meaningful, and even what aspects of experience (material, affective, symbolic, social) are allowed to appear are all prefigured by the initial act of rendering.

To demonstrate that this reductive dynamic is not unique to philosophy but operates across divergent methods, it is necessary to examine how similar stabilization-through-reduction appears in other prominent approaches to technology. In the framework of cultural techniques as reconstructed by Geoghegan, for instance, reduction functions through a procedural foregrounding that stabilizes technology by redescribing it as a sequence of technical operations said to "historically and logically precede"[4] media objects or concepts. This move promises to return analysis to a more elemental level, yet it readily slips into a further reduction to text when the most primary technical form is assumed to be a formalized symbolic system. As Geoghegan notes, "Cultural theorists at the Humboldt University of Berlin (e.g. Christian Kassung, Sybille Kramer and Thomas Macho) identify cultural techniques with rigorous and formalized symbolic systems, such as reading, writing, mathematics, music, and imagery (see Kassung and Macho, 2013; Kramer and Bredekamp, 2008; Macho, 2013)".[5] Even when the method refrains from explicitly privileging a symbolic substrate, the technical is often quietly displaced by the symbolic through its conflation with "the rules of selection, storage, and transmission that characterize a given system of mediation, including the formal structures that compose and constrict this process".[6] The result is a subtle but consequential substitution: material and affective technical conditions give way to symbolic-operational rules. A paradigmatic instance of this slippage appears in Kittler's well-known conflation of the machine with the organization of the psychic apparatus (thus the 1997 "The World of the Symbolic—A World of the Machine").

[4]Bernhard Siegert, "The Map Is the Territory," *Radical Philosophy* 169 (2011): 15.

[5]Bernard Dionysius Geoghegan, "After Kittler: On the Cultural Techniques of Recent German Media Theory," *Theory, Culture & Society* 30, no. 6 (2013): 67.

[6]Geoghegan, "After Kittler: On the Cultural Techniques of Recent German Media Theory," 69.

In the early history of Science and Technology Studies, the movement retrospectively labeled the "turn to technology"[7] did not emerge as a centering of technology as a primary object of inquiry; instead, it appeared as a reorientation in which technological artifacts, instruments, and infrastructures were selectively taken up because they promised to resolve or illuminate problems that had already been defined within social theory. Technology entered the analytic frame not on its own terms, but as a corrective lens through which long-standing questions about the organization of social order and the sources of epistemic authority could be reexamined with greater material specificity. This framing carried a quiet but decisive reduction: the technological was stabilized only insofar as it could be made to speak to concerns that preceded and exceeded it. What resulted was a distinctive mode of engagement in which the material presence of technologies mattered less for their own operational logic than for the ways they could be shown to participate in the negotiation and maintenance of social relations.

This orientation becomes especially visible in the canonical questions that have structured much of STS scholarship: How are technologies socially constructed through ongoing negotiations among heterogeneous networks of human and nonhuman actors?[8] How do material artifacts and systems embody and enforce particular social norms and expectations, thereby enabling some forms of action while constraining or rendering others impossible?[9] And how do standards, classifications, and large technical infrastructures quietly organize both everyday social practices and the production of knowledge, all while receding into the background of ordinary experience?[10] Through these lines of inquiry, STS has developed powerful tools for mapping sociomaterial entanglements and revealing the invisible work of infrastructure. At the same time, however, this approach has consistently rendered technology legible primarily as a vehicle or embodiment of social and epistemic dynamics. The technology itself—its specific modes of operation, its material affordances beyond their social translation, and its capacity to generate effects that cannot be fully captured in terms of construction or order—tends to remain secondary. In

[7] Steve Woolgar, "The Turn to Technology in Social Studies of Science," *Science, Technology, & Human Values* 16, no. 1 (1991): 20–50.

[8] Trevor J. Pinch and Wiebe E. Bijker, "The Social Construction of Facts and Artefacts: Or How the Sociology of Science and the Sociology of Technology Might Benefit Each Other," *Social Studies of Science* 14, no. 3 (1984): 399–441.

[9] Langdon Winner, "Do Artifacts Have Politics?," *Daedalus* 109, no. 1 (1980): 121–136.

[10] Geoffrey C. Bowker and Susan Leigh Star, *Sorting Things Out: Classification and Its Consequences* (The MIT Press, 1999), https://direct.mit.edu/books/monograph/4738/Sorting-Things-OutClassification-and-Its.

this way, the "turn to technology" reveals a deeper pattern: even as STS foregrounds the inseparability of the social and the material, it often stabilizes the technological object through a reduction that subordinates its own specificity to the illumination of pre-existing social-theoretical problems.

Reduction, as we have seen by now, is a procedural condition for stabilizing technology as an object of discourse. Its particular protocol—what is foregrounded, what is excluded, and at what scale analysis proceeds—appears to be shaped by the distinctive focus and commitments of each methodology. In light of the object, this observation immediately raises a further question: if the form of reduction depends on what aspects of a technology can be rendered analyzable within a given discursive frame, might the operational characteristics of the technology itself influence the kinds of reduction it invites or affords? In other words, do technologies differ in the ways they operate, produce outputs, and structure information, such that some forms of abstraction or representation become more natural, compelling, seemingly inevitable, than others?

3.2 Redefining Affordances

The material structures of large language models do not merely invite textual reduction; they actively enforce it as a constitutive operational necessity. This enforcement arises directly from the technical limitations inherent to any attempt to render natural language computable. Because exhaustive semantic modeling remains intractable, systems must rely on statistical approximations operating at scales and granularities that detach surface linguistic performance from the underlying machinery. The result is not an accidental opacity awaiting external interpretation, but a designed architecture in which fluent textual output is sustained precisely by rendering the generative processes inaccessible to linguistic or symbolic scrutiny. Reduction ceases to be a methodological habit of media studies or STS and becomes instead an affordance materially instantiated by the technology itself.

At the core of this dynamic lies the unavoidable translation problem that defines language technologies, one of translation between human and machine. Natural language cannot enter computational processes in its raw, continuous form: it must undergo a fundamental conversion into discrete, numerically manipulable units optimized for matrix operations and probabilistic prediction. These units, however, cannot preserve stable correspondence to linguistic categories such as words, morphemes, characters; the constraints of vocabulary size, sequence length,

and training efficiency compel a decomposition into subword fragments whose statistical utility supplants any semantic coherence. What emerges is a layer of representation that functions effectively for generation and pattern completion yet remains constitutively non-linguistic. The model produces coherent discourse not because it manipulates meaning, but because it has learned to navigate distributional regularities at a level beneath linguistic structure. This sub-linguistic substrate is the condition of possibility for the surface fluency that defines the system's utility.

Consequently, any attempt to engage LLMs primarily through textual analysis or discursive framing encounters a technology that has already performed the reduction in advance. The interface delivers outputs that reward interpretation as meaningful communication, while the actual operations—nonlinear transformations across vast parameter spaces—operate according to logics indifferent to human categories of understanding. This is not a failure of explainability but the successful realization of an engineering paradigm in which performance at the linguistic surface is decoupled from interpretability at the operational core. The technical limitations of translation thus materialize as an architectural feature: language is made computable only by stripping away the very dimensions that would allow direct symbolic access to its processing.

This enforced technesis carries significant epistemic implications for fields concerned with technology. When scholars stabilize LLMs through familiar lenses, they risk aligning with the very logic the systems embed. Such approaches treat the textual interface as the primary site of analysis, thereby reproducing the separation the architecture already enforces. The challenge is therefore not to overcome opacity through better metaphors or more nuanced social contextualization, but to recognize how certain material configurations systematically privilege textual legibility while rendering non-discursive dimensions illegible. LLMs structurally solicit and reward reduction to text, shaping the kinds of questions and interpretations that appear natural or productive.

Most broadly, this thesis suggests that technologies of language generation invite a reevaluation of how we conceptualize the relationship between material operation and symbolic representation. The persistent tendency to evaluate intelligence or capability through textual performance—from early thought experiments to contemporary benchmarking—reflects not a cultural bias, as it often is posed, but a technical affordance coming from the practical necessities of making language tractable for computation. Or: where traditional reductions in theory served as

necessary simplifications, here the simplification is engineered into the artifact. Recognizing this dynamic opens the possibility for forms of inquiry that resist the enforced reduction, attending instead to the generative tension between the inseparable computational substrate and linguistic surface. By holding this tension open, we can begin to engage with the distinctive material reality of natural language generation technologies without being captured by the (textual) appearances they so effectively sustain.

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